Construction manual

Version 1 — powered by a 9 V battery

Part	Support with light for the pregoscope (telescopic-tube microscope, 40x objective)
Units	millimetres (mm) · first-angle projection
Tolerance	±0.2 mm (ISO 2768-m; PLA, 0.2 mm layer)
Origin	on the tube axis, z = 0 at the top face of the lid (floor of the box), +z up, +x to the right seen from the front, +y toward the column
Estimated time	160 min of assembly, not counting printing

Status: pre-calibration

The dimensions are the design values and have not been checked on printed parts yet. Marked (*) in the drawings.

- Outer tube: the STEP said 30 mm and the caliper 33.5 mm. The 3D and the clamp bore use 33.5 mm; confirm with jig G1.
- Objective: 38 mm total length (manufacturer data) and 35 mm below the inner tube (measured); the real working distance (0.4 to 0.8 mm) depends on the model and must be checked while focusing.
- Cut-outs of the switch, jack and knob, LED flange, slide and screws: confirm with jig G2; the 9 V battery with G3.
- The traces and the solder mask of the boards passed an automatic check but have not been looked at in PCB software.
- The Chinese and English text was written without a technical translator and needs review by a native speaker.

Revisions

Rev.	Date	Description
A	2026-11-03	First issue, pre-calibration

Before you start

Tools and materials

- FDM 3D printer, bed of at least 300 mm, PLA filament
- 30 to 40 W soldering iron, 0.8 mm solder, wire stripper
- Multimeter (DC voltage and current)
- Hot-glue gun, heat gun or lighter for the heat-shrink
- Small Phillips screwdriver; 3 mm drill bit; fine sandpaper; caliper

Safety

- Soldering iron at about 300 °C: use the stand, ventilate the room, never touch the tip.
- Hot glue and heat-shrink burn: let them cool before handling.
- Check the polarity before powering up (LED, diode, battery, positive-centre plug). A reversed connection burns components.
- Use only the 12 V supply listed (v2) and a new 9 V alkaline battery (v1). Do not use rechargeable lithium batteries.
- Do not overtighten screws in PLA and do not force the glass slide.

Printing

PLA · 0.2 mm · 3 perimeters · 20 to 30 % infill. Dark walls help to reduce stray light.

ID	Part	Orientation
M1	Support: light box, column and arm	file -impressao, lying on its back; supports only inside the box
M2	Tube clamp	file -impressao, opening upward; no supports
M3	Bottom lid	already flat; no supports
M4	Diffuser	flat, natural or white PLA, 1 to 2 layers; no supports

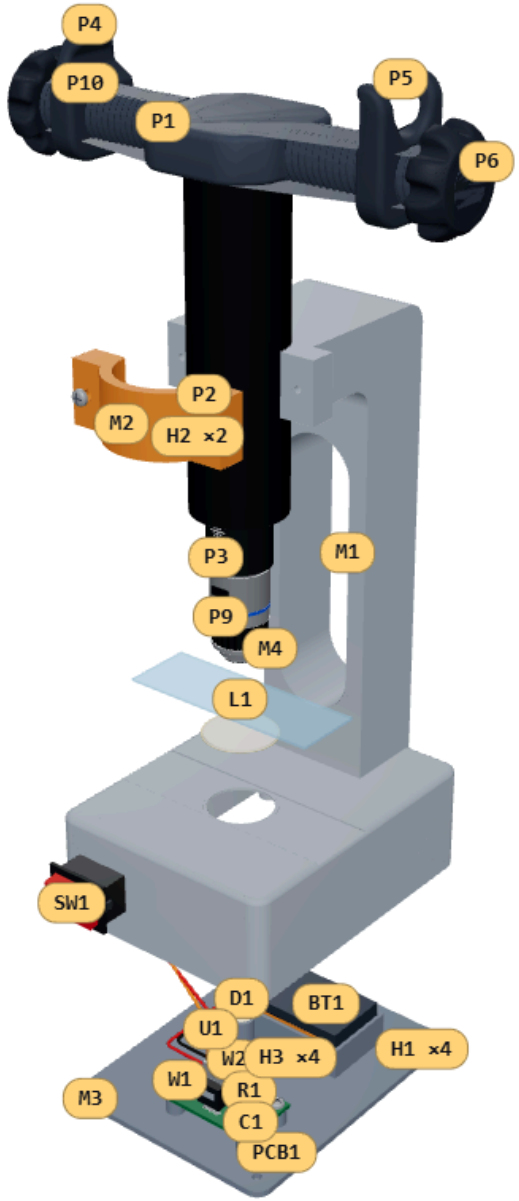
Files that come with this manual

File	Content
`mecanica/v1-pilha/suporte-base-coluna-impressao.stl`	M1 support, in the print orientation
`mecanica/v1-pilha/suporte-base-coluna.stl`	M1 support, in the assembly position
`mecanica/comum/abracadeira-tubo-impressao.stl`	M2 clamp, in the print orientation
`mecanica/v1-pilha/tampa-fundo.stl`	M3 bottom lid
`mecanica/comum/difusor.stl`	M4 diffuser

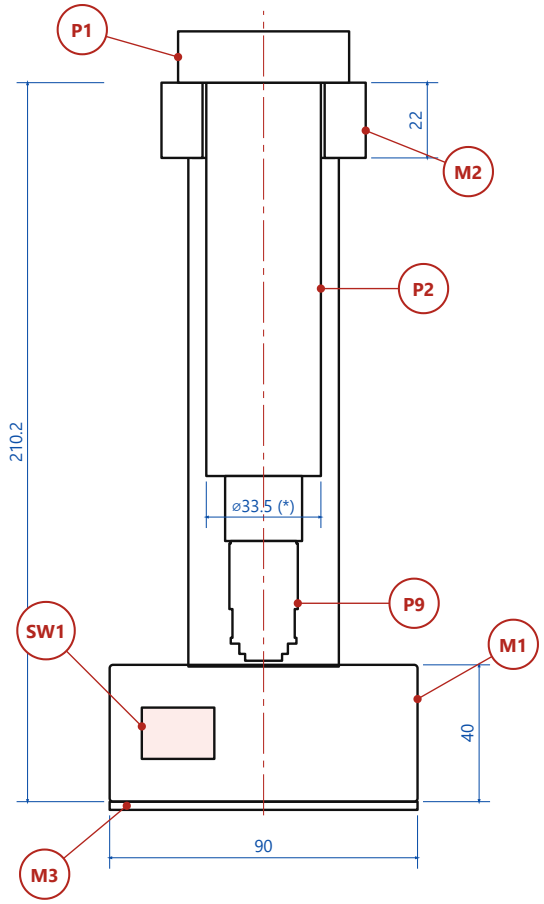
File	Content
`eletronica/v1-pilha/esquematico.svg`	Circuit schematic
`eletronica/v1-pilha/gerber/`	Gerber files of the board PCB1 (fabrication)
`eletronica/v1-pilha/bom.csv`	Component list of the board
`eletronica/v1-pilha/pick-and-place.csv`	Component placement coordinates
`lista-de-compras-v1.csv`	Shopping list (price to be filled in)
`mecanica/calibracao/`	Jigs G1 to G3 (print them first)

Order of the document: exploded view · 1 technical drawings · 2 positions · 3 bill of materials · 4 wiring · 5 assembly · 6 test · 7 calibration · 8 inspection.

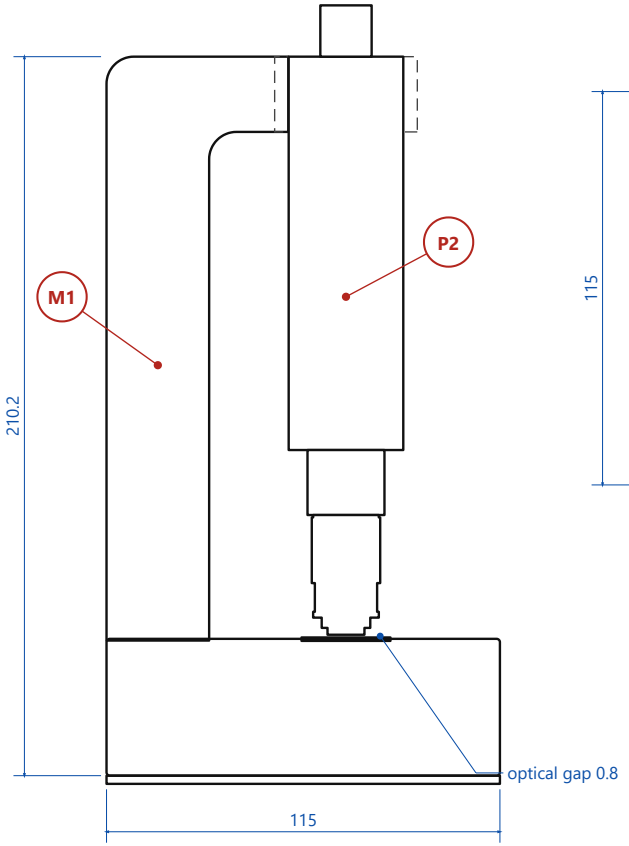
Exploded view



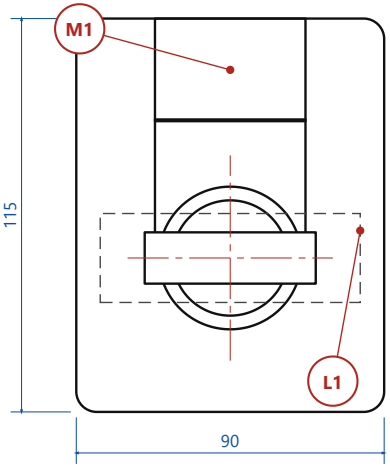
Every part pulled apart with its ID from the bill of materials; “×n” is the quantity.



Front view
1:2



Left view
1:2



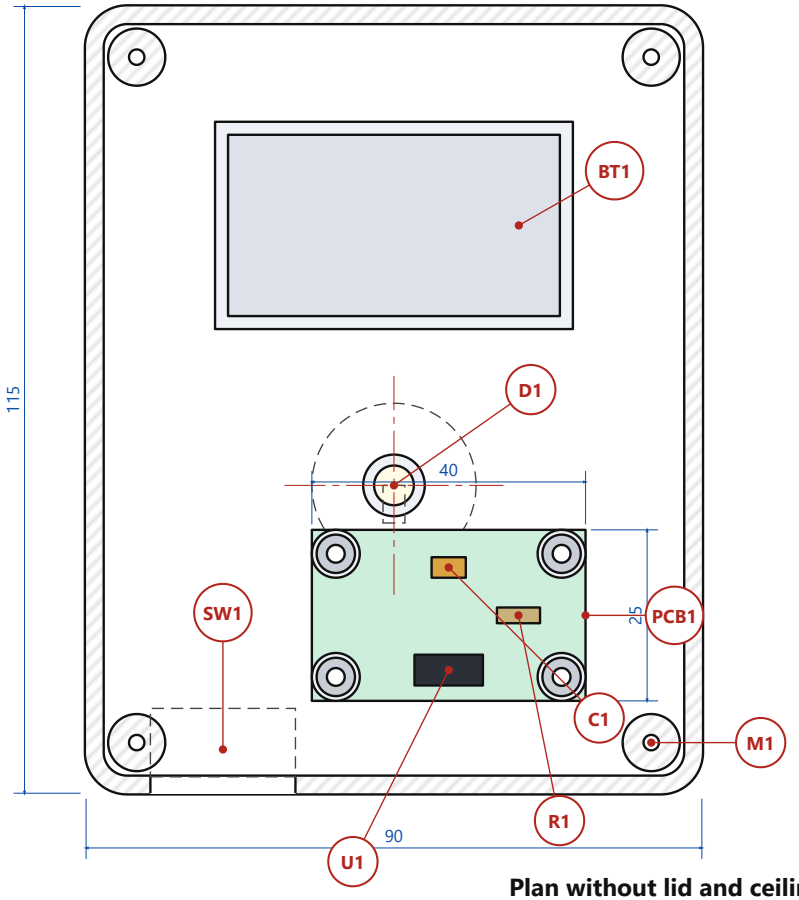
Top view
1:2

Tube ø33.5 (*) · sleeve ø22.5 · objective ø20 × 35 · lens at z = 41.2 · gap 0.8

General notes

1. Dimensions in mm. General tolerance ±0.2 mm (ISO 2768-m). Fit clearances follow jigs G1 to G3.
2. PLA, 0.2 mm layer, 3 perimeters, 20 to 30 % infill. Print orientation: see the cover.
3. Dimensions marked (*) depend on the physical calibration and may change in revision B.

Title: **General assembly**			Code: **DT-01-v1**
Part: Assembly	Material: PLA	Scale: 1:2	Sheet: 1 / 9
Units: mm	Tolerance: ±0.2 (ISO 2768-m)	Revision: A · 2026-11-03	Version: v1 9 V battery
Project: Pregoscope: support with light		Projection: 1st angle	



Plan without lid and ceiling

1:1

Centre position of each part (mm)

ID	X	Y	Z	Size
D1	0	0	31.3	5.8 × 5.8 × 8.6
PCB1	8	-19	6.8	40 × 25 × 1.6
U1	8	-27	18.4	10 × 4.5 × 15.5
R1	18.2	-19	9	6.3 × 2.3 × 2.3
C1	8	-12	11.1	5 × 2.6 × 5.4
SW1	-25	-43.8	20	21.2 × 10 × 15
H3	8	-19	7.6	5.6 × 5.6 × 2
W1	0	-8	8	3 × 20 × 3
W2	0	-8	3	3 × 10 × 3
BT1	0	38	8.8	48.5 × 26.5 × 17.5

Origin: tube axis, z = 0 at the floor of the box.

PCB1: 40 × 25 × 1.6 mm, top face at z = 7.6.

Wires: see DT-03 and the wiring order (section 4).

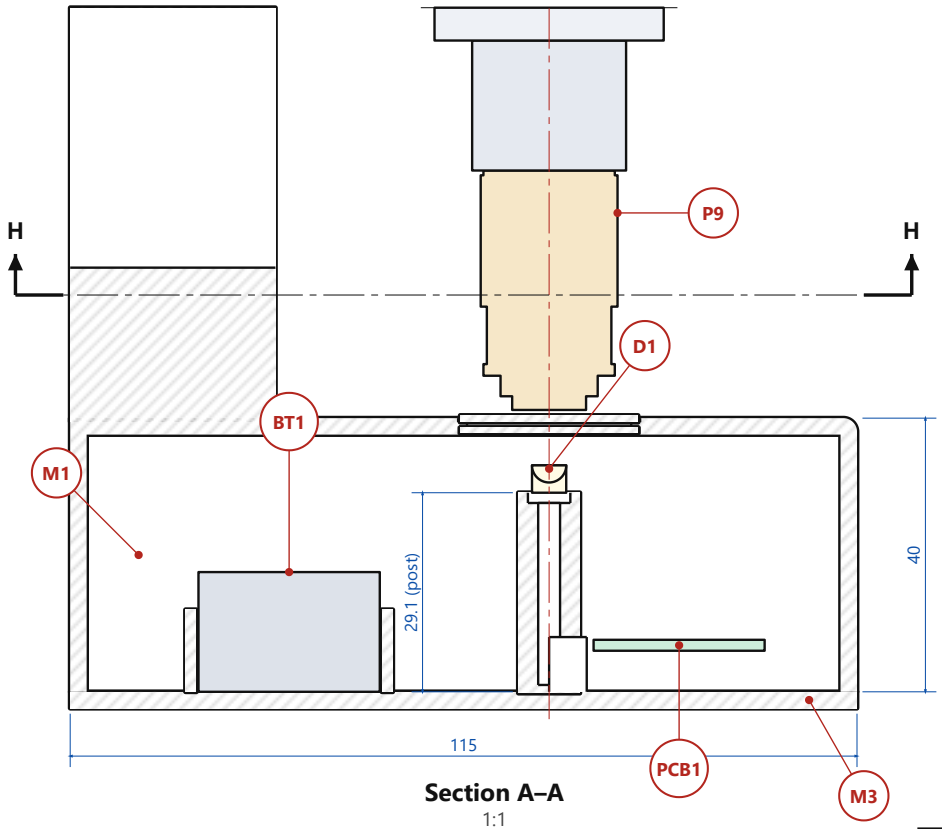
General notes

1. Dimensions in mm. General tolerance ±0.2 mm (ISO 2768-m). Fit clearances follow jigs G1 to G3.

2. PLA, 0.2 mm layer, 3 perimeters, 20 to 30 % infill. Print orientation: see the cover.

3. Dimensions marked (*) depend on the physical calibration and may change in revision B.

Title: **Master plan: inside the box (horizontal section z = 20)**			Code: **DT-02-v1**
Part: Assembly	Material: PLA	Scale: 1:1	Sheet: 2 / 9
Units: mm	Tolerance: ±0.2 (ISO 2768-m)	Revision: A · 2026-11-03	Version: v1 9 V battery
Project: Pregoscope: support with light		Projection: 1st angle	

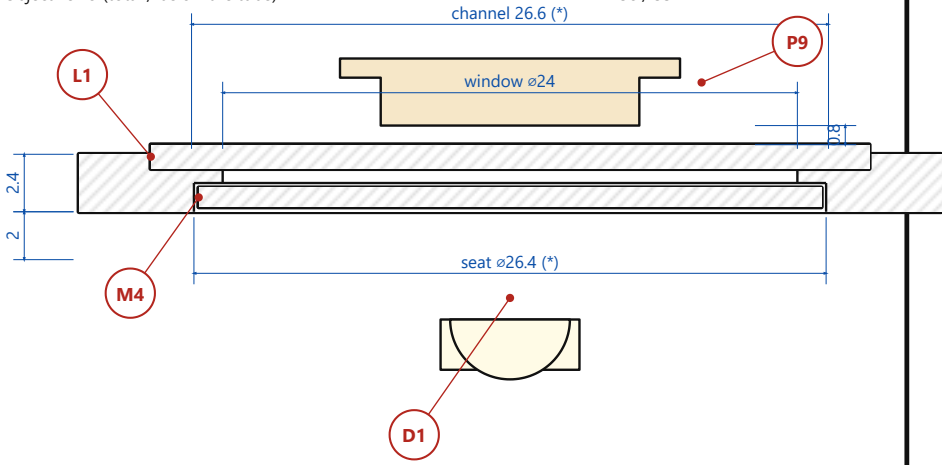


Section A–A

1:1

Optical stack (bottom to top)

LED D1 (top)	z = 35.6
Diffuser M4	z = 37.8–38.6
Top face of the ceiling	z = 40
Slide L1 (top)	z = 40.4
Lens of objective P9	z = 41.2
Working distance	0.8 mm
Objective P9 (total / below the tube)	38 / 35 mm



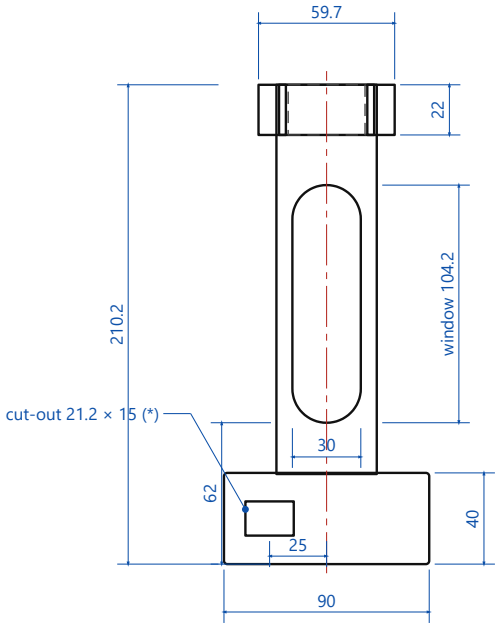
Detail H: optical stack

3.5:1

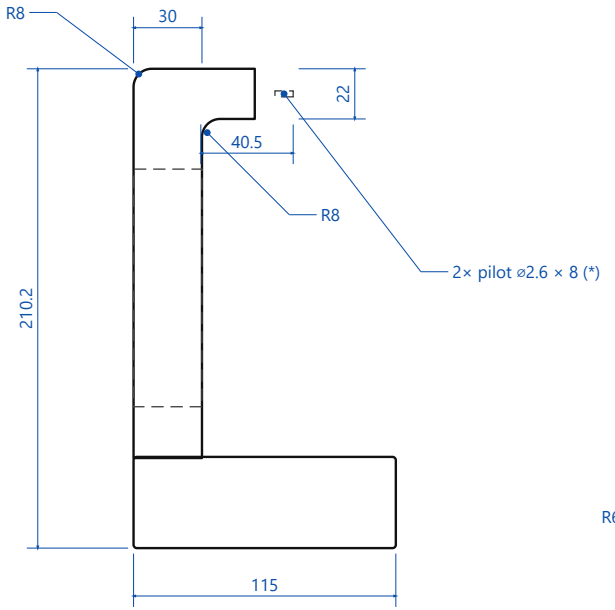
General notes

1. Dimensions in mm. General tolerance ±0.2 mm (ISO 2768-m). Fit clearances follow jigs G1 to G3.
2. PLA, 0.2 mm layer, 3 perimeters, 20 to 30 % infill. Print orientation: see the cover.
3. Dimensions marked (*) depend on the physical calibration and may change in revision B.

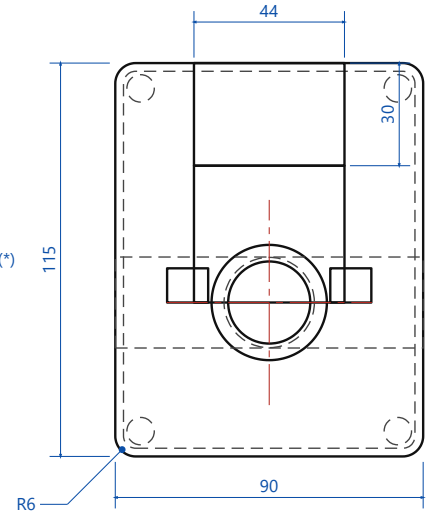
Title: **Master plan: section A–A (x = 0), optical stack**			Code: **DT-03-v1**
Part: Assembly	Material: PLA	Scale: 1:1 · 3.5:1	Sheet: 3 / 9
Units: mm	Tolerance: ±0.2 (ISO 2768-m)	Revision: A · 2026-11-03	Version: v1 9 V battery
Project: Pregoscope: support with light		Projection: 1st angle	



Front view
1:3



Left view
1:3



Top view
1:2

Main cut-outs

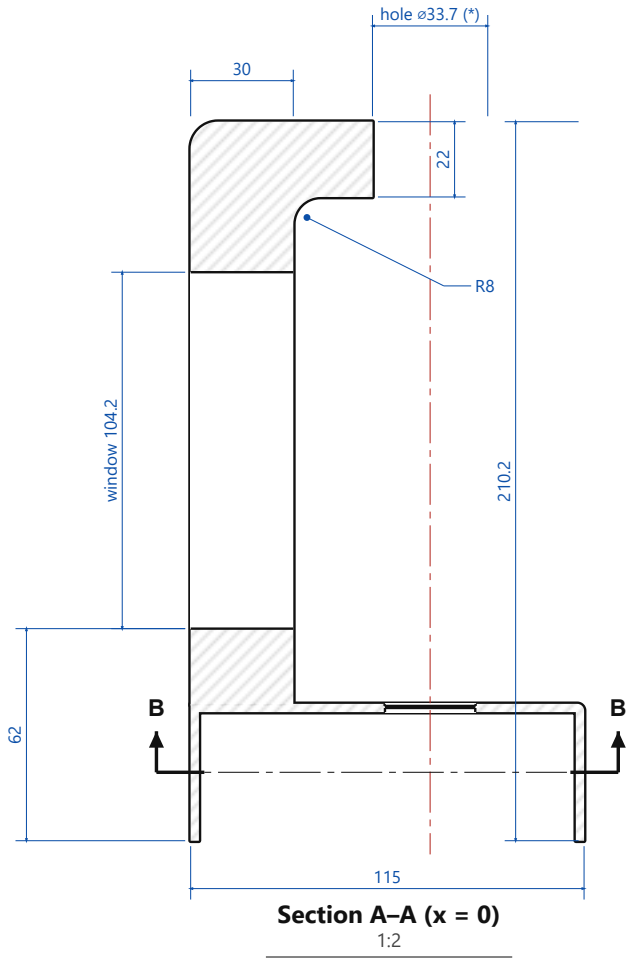
LED window ⌀24 · diffuser seat ⌀26.4 (*) · slide channel 26.6 (*) × 0.6

ring bore ⌀33.7 (*) · column window 30 × 104.2 · cut-out SW1 21.2 × 15 (*)

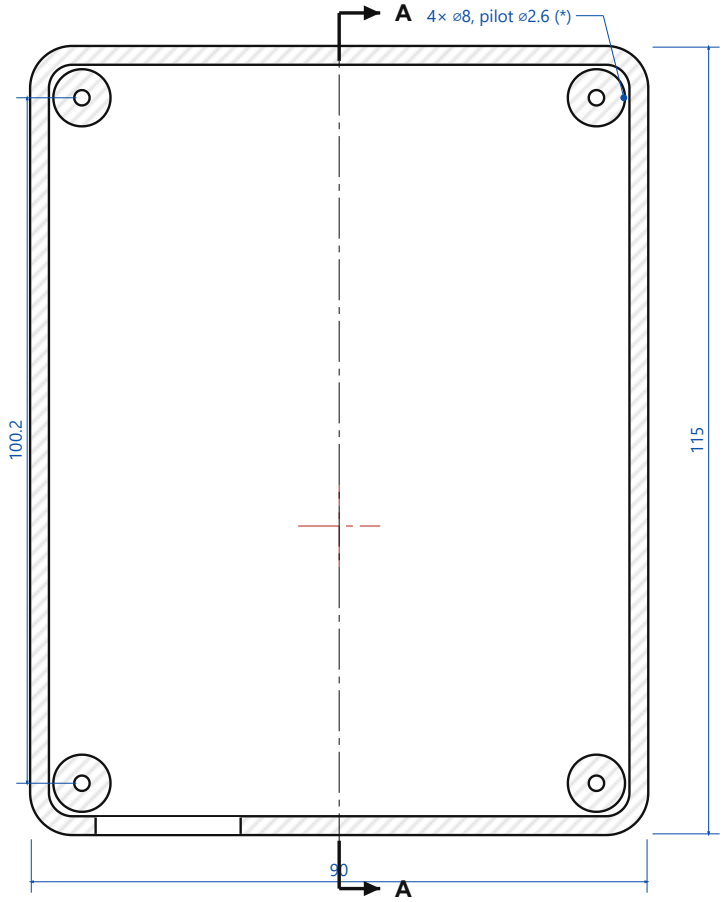
General notes

1. Dimensions in mm. General tolerance ±0.2 mm (ISO 2768-m). Fit clearances follow jigs G1 to G3.
2. PLA, 0.2 mm layer, 3 perimeters, 20 to 30 % infill. Print orientation: see the cover.
3. Dimensions marked (*) depend on the physical calibration and may change in revision B.

Title		Code	
M1 Support: external views M1		**DT-04-v1**	
Part: M1	Material: PLA	Scale: 1:3 · 1:2	Sheet: 4 / 9
Units: mm	Tolerance: ±0.2 (ISO 2768-m)	Revision: A · 2026-11-03	Version: v1 9 V battery
Project: Pregoscope: support with light		Projection: 1st angle	



Section A–A (x = 0)
1:2



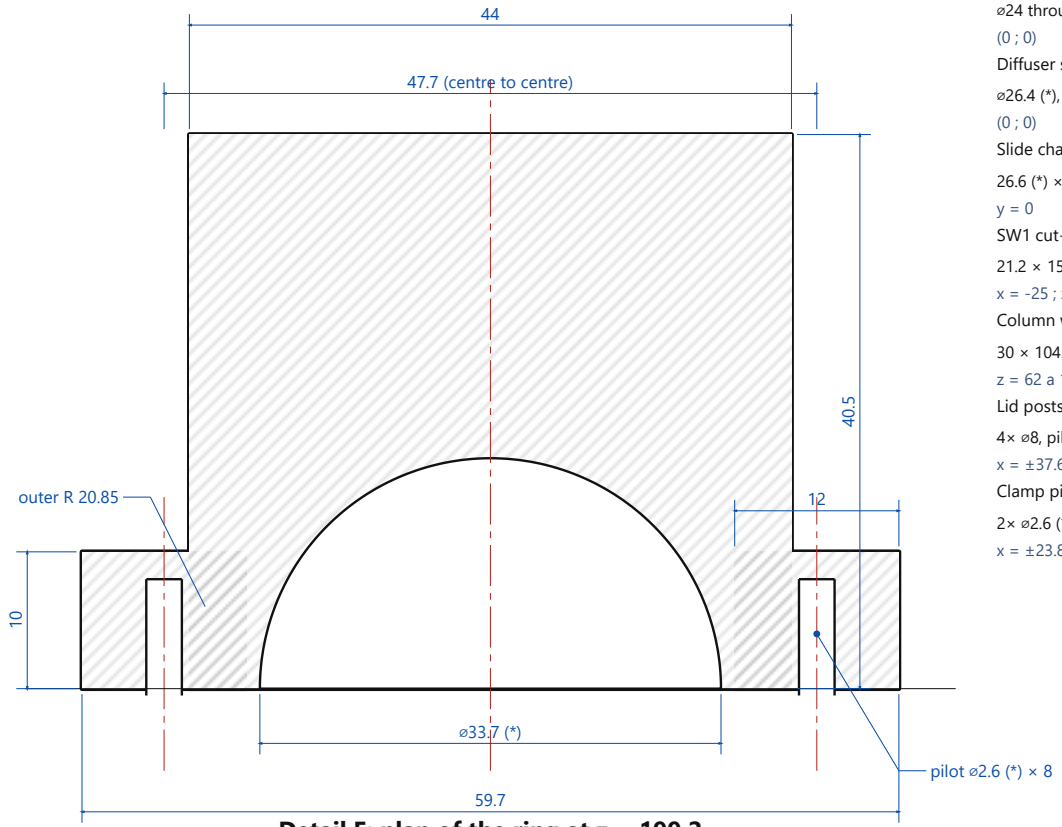
Section B–B (z = 20)
1:1

Ceiling, diffuser seat and channel: detail H (DT-03) and detail G (DT-09).

General notes

1. Dimensions in mm. General tolerance ±0.2 mm (ISO 2768-m). Fit clearances follow jigs G1 to G3.
2. PLA, 0.2 mm layer, 3 perimeters, 20 to 30 % infill. Print orientation: see the cover.
3. Dimensions marked (*) depend on the physical calibration and may change in revision B.

Title			Code
M1 Support: sections A–A and B–B M1			**DT-05-v1**
Part: M1	Material: PLA	Scale: 1:2 · 1:1	Sheet: 5 / 9
Units: mm	Tolerance: ±0.2 (ISO 2768-m)	Revision: A · 2026-11-03	Version: v1 9 V battery
Project: Pregoscope: support with light		Projection: 1st angle	



Detail E: plan of the ring at z = 199.2

2:1

Table of cut-outs and holes of M1

LED window

⌀24 through the ceiling

(0 ; 0)

Diffuser seat

⌀26.4 (*), from inside the ceiling

(0 ; 0)

Slide channel

26.6 (*) × 0.6 deep, along x

y = 0

SW1 cut-out

21.2 × 15 (*), front wall

x = -25 ; z = 20

Column window

30 × 104.2, through along y

z = 62 a 166.2

Lid posts

4× ⌀8, pilot ⌀2.6 (*)

x = ±37.6 ; y = -37.6 e 62.6

Clamp pilot hole

2× ⌀2.6 (*) × 8 deep, along y

x = ±23.85 ; z = 199.2

(*) to be confirmed with jigs G1 to G3.

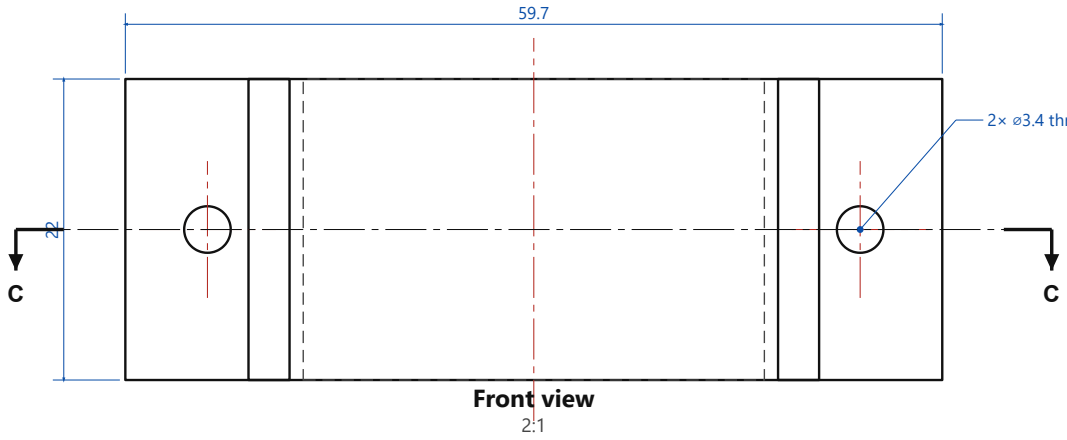
General notes

1. Dimensions in mm. General tolerance ±0.2 mm (ISO 2768-m). Fit clearances follow jigs G1 to G3.

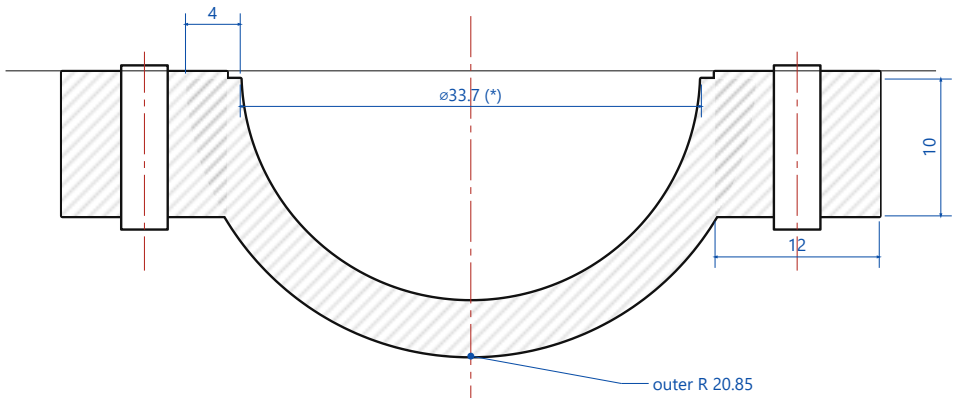
2. PLA, 0.2 mm layer, 3 perimeters, 20 to 30 % infill. Print orientation: see the cover.

3. Dimensions marked (*) depend on the physical calibration and may change in revision B.

Title		Code	
M1: ring detail and cut-outs M1		**DT-06-v1**	
Part: M1	Material: PLA	Scale: 2:1	Sheet: 6 / 9
Units: mm	Tolerance: ±0.2 (ISO 2768-m)	Revision: A · 2026-11-03	Version: v1 9 V battery
Project: Pregoscope: support with light		Projection: 1st angle	



Front view
2:1



Section C–C (z = 199.2)
2:1

Print orientation

File abracadeira-tubo-impressao.stl (opens upward), no supports.

Assembly

2× screw M3 × 16 (H2) go through the ears and enter 8 mm into the pilot hole ø2.6 of M1.

Tube bore

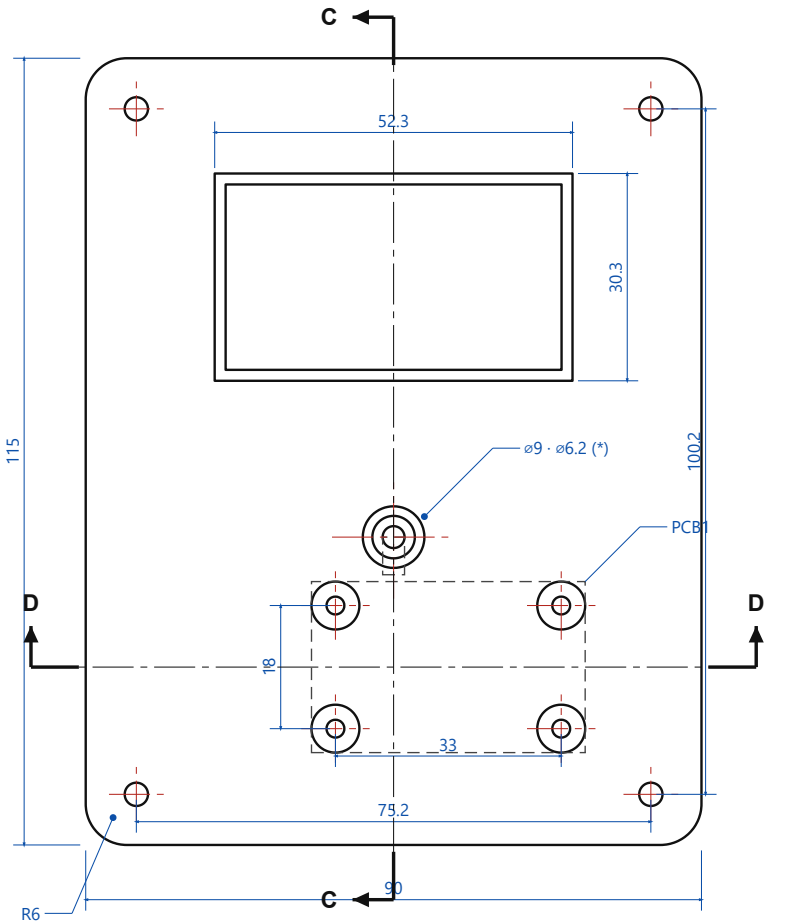
ø33.7 (*) = measured tube 33.5 + gap 0.2.

Confirm with jig G1 before printing in series.

General notes

1. Dimensions in mm. General tolerance ±0.2 mm (ISO 2768-m). Fit clearances follow jigs G1 to G3.
2. PLA, 0.2 mm layer, 3 perimeters, 20 to 30 % infill. Print orientation: see the cover.
3. Dimensions marked (*) depend on the physical calibration and may change in revision B.

Title		Code	
M2 Tube clamp (removable half) M2		**DT-07-v1**	
Part: M2	Material: PLA	Scale: 2:1	Sheet: 7 / 9
Units: mm	Tolerance: ±0.2 (ISO 2768-m)	Revision: A · 2026-11-03	Version: v1 9 V battery
Project: Pregoscope: support with light		Projection: 1st angle	



Top view (print orientation)
1:1

Lid holes (H1)
4× ø3.4 — x = ±37.6 ; y = -37.6 / 62.6

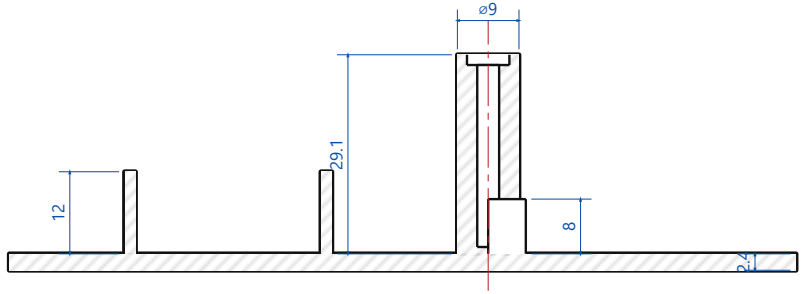
Board standoffs (H3)
4× ø7, pilot ø2.6 (*), h = 6 — x = -8.5 / 24.5 ; y = -28 / -10

LED post (D1)
ø9, h = 29.1 — (0 ; 0)

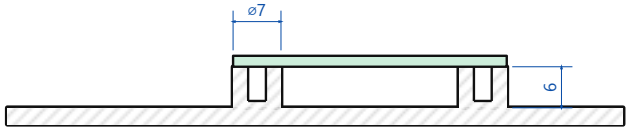
Flange seat
ø6.2 (*) × 1.5 — (0 ; 0)

Leg hole
ø3.2 blind + slot 3.2 × 5.5 × 8 — slot toward −y

Battery cradle
52.3 × 30.3, h = 12, wall 1.6 — centre (0 ; 38)



Section C–C (x = 0)
1:1

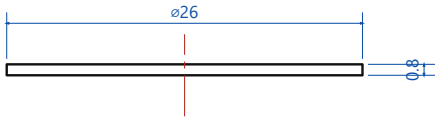


Section D–D (y = -19)
1:1

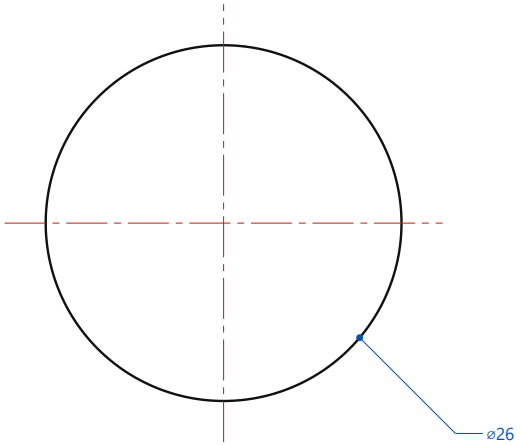
General notes

1. Dimensions in mm. General tolerance ±0.2 mm (ISO 2768-m). Fit clearances follow jigs G1 to G3.
2. PLA, 0.2 mm layer, 3 perimeters, 20 to 30 % infill. Print orientation: see the cover.
3. Dimensions marked (*) depend on the physical calibration and may change in revision B.

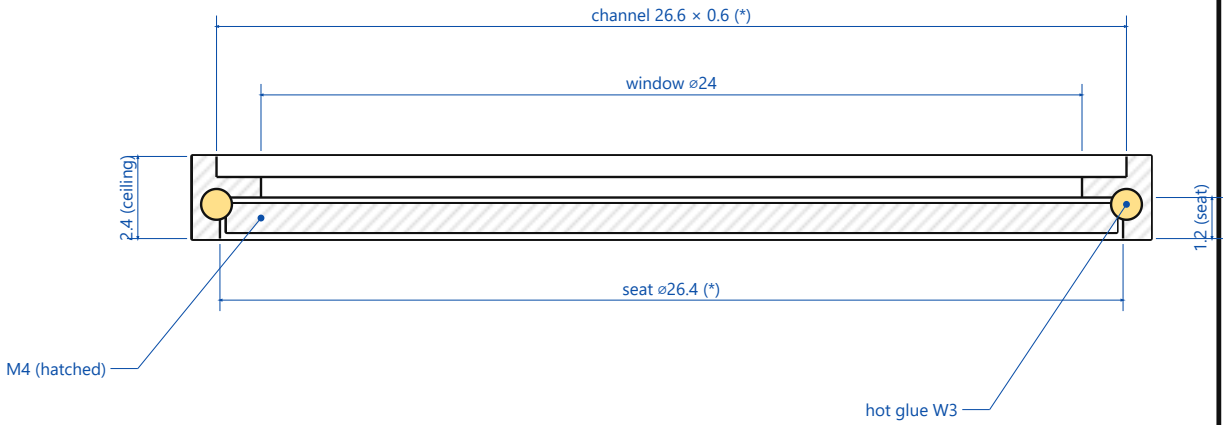
Title: **M3 Bottom lid M3**			Code: **DT-08-v1**
Part: M3	Material: PLA	Scale: 1:1	Sheet: 8 / 9
Units: mm	Tolerance: ±0.2 (ISO 2768-m)	Revision: A · 2026-11-03	Version: v1 9 V battery
Project: Pregoscope: support with light		Projection: 1st angle	



Front view
2:1



Top view
2:1



Detail G: ceiling of M1 in section
5:1

Material and colour

Natural or white PLA, 1 to 2 layers of 0.2 mm (≈ 0.8 mm). No supports.

General notes

1. Dimensions in mm. General tolerance ±0.2 mm (ISO 2768-m). Fit clearances follow jigs G1 to G3.
2. PLA, 0.2 mm layer, 3 perimeters, 20 to 30 % infill. Print orientation: see the cover.
3. Dimensions marked (*) depend on the physical calibration and may change in revision B.

Title: **M4 Diffuser and its seat in the ceiling of M1 M4**			Code: **DT-09-v1**
Part: M4	Material: PLA	Scale: 2:1 · 5:1	Sheet: 9 / 9
Units: mm	Tolerance: ±0.2 (ISO 2768-m)	Revision: A · 2026-11-03	Version: v1 9 V battery
Project: Pregoscope: support with light		Projection: 1st angle	

2. Position of each part

Centre of the part in the coordinate system of the cover (mm). “Inside” means the part is inside the closed box.

ID	Name	X	Y	Z	Size X × Y × Z	Fixing	Inside
M1	Support: light box, column and arm	0	12.5	105.1	90 × 115 × 210.2	standing on the table	no
M3	Bottom lid	0	12.5	-1.2	90 × 115 × 2.4	4 screws H1 at the corners	yes
M4	Diffuser	0	0	38.2	26 × 26 × 0.8	hot glue (W3) in the ceiling seat	yes
M2	Tube clamp	0	-10.7	199.2	59.7 × 20.3 × 22	2 screws H2 in the ears	no
L1	Microscope slide	0	0	39.9	76 × 26 × 1	slid into the channel from the side	no
D1	White 5 mm LED (diffused)	0	0	31.3	5.8 × 5.8 × 8.6	seat of the LED post on the lid	yes
PCB1	Driver board	8	-19	6.8	40 × 25 × 1.6	4 screws H3 on the 6 mm standoffs	yes
U1	LM317T adjustable regulator	8	-27	18.4	10 × 4.5 × 15.5	soldered to PCB1, tab facing back	yes
R1	Resistor R1 (sets the current)	18.2	-19	9	6.3 × 2.3 × 2.3	soldered to PCB1	yes
C1	Ceramic capacitor 100 nF	8	-12	11.1	5 × 2.6 × 5.4	soldered to PCB1, next to U1	yes
SW1	Rocker switch KCD1-101	-25	-43.8	20	21.2 × 10 × 15	front-wall cut-out, snap tabs	no
H1	M3 × 10 self-tapping screw	0	0	-1.2	5.6 × 5.6 × 2	4 lid holes (see the table below)	yes
H2	M3 × 16 self-tapping screw	0	-12.8	199.2	54.4 × 16 × 5.6	2 clamp ears, Y axis (see the table below)	no
H3	M3 × 8 self-tapping screw	8	-19	7.6	5.6 × 5.6 × 2	4 PCB1 holes (see the table below)	yes
W1	Flexible wire 24 AWG	0	-8	8	3 × 20 × 3	from the LED to PCB1 through the side slot of the LED post	yes
W2	Heat-shrink tube ⌀2-3 mm	0	-8	3	3 × 10 × 3	over the two LED solder joints	yes
W3	Hot glue and double-sided tape	0	0	38	26 × 26 × 1	bead around the diffuser edge	yes
P9	Objective 40x / 0.65	0	0	58.7	20 × 20 × 35	screwed to the inner tube of the pregoscope	no
P2	Outer tube	0	0	152.7	33.5 × 33.5 × 115	in the column ring, held by M2	no
P1	Head: plate and threaded rod	0	0	217.7	50 × 50 × 15	resting on top of the ring	no
BT1	9 V battery (6F22) with snap clip	0	38	8.8	48.5 × 26.5 × 17.5	lid cradle, snap clip toward the left	yes

Screws, one by one

ID	Screw	X	Y	Z
H1	M3 × 10 self-tapping screw	-37.6	-37.6	-2.4
H1	M3 × 10 self-tapping screw	-37.6	62.6	-2.4
H1	M3 × 10 self-tapping screw	37.6	-37.6	-2.4
H1	M3 × 10 self-tapping screw	37.6	62.6	-2.4
H3	M3 × 8 self-tapping screw	-8.5	-28	6
H3	M3 × 8 self-tapping screw	-8.5	-10	6
H3	M3 × 8 self-tapping screw	24.5	-28	6
H3	M3 × 8 self-tapping screw	24.5	-10	6
H2	M3 × 16 self-tapping screw	-23.8	-12.8	199.2
H2	M3 × 16 self-tapping screw	23.9	-12.8	199.2

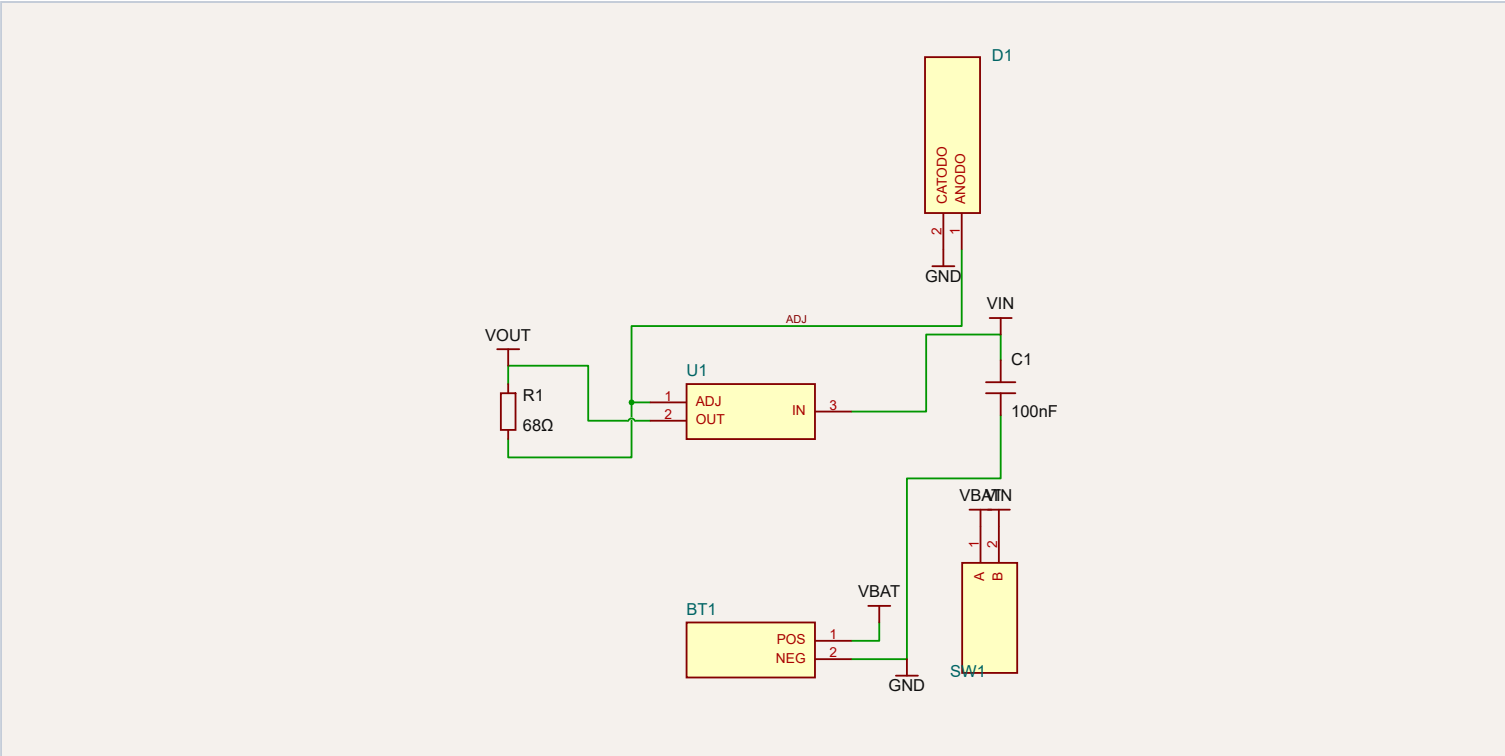
3. Bill of materials (BOM)

ID	Name	Qty	Standard size	Type
M1	Support: light box, column and arm	1	PLA, base 90 × 115 × 40 mm, 210,2 mm de altura, cantos de 6 mm, janela de alívio de 30 mm na coluna	printed
M2	Tube clamp	1	Anel de 22 mm de altura, furo de 33,7 mm, dois parafusos M3 × 16	printed
M3	Bottom lid	1	Placa de 2,4 mm com porta-LED; v1 leva o berço da pilha, v2 leva 4 suportes da placa do dimmer	printed
M4	Diffuser	1	Disco de ⌀ 26 × 0,8 mm em PLA natural ou branco	printed
P1	Head: plate and threaded rod	1	50 × 15 mm, haste de 170 mm	supplied (pregoscope)
P2	Outer tube	1	tube ⌀ 33.5 mm × 115 mm (measured)	supplied (pregoscope)
P3	Inner tube (telescopic)	1	tube ⌀ 22.5 mm, pulls out up to 40 mm, printed mm scale	supplied (pregoscope)
P4	Left jaw	1	10 mm de espessura	supplied (pregoscope)
P5	Right jaw	1	10 mm de espessura	supplied (pregoscope)
P6	Right knob	1	⌀ 34 mm	supplied (pregoscope)
P10	Left knob (mirror of P6)	1	⌀ 34 mm, igual à P6	supplied (pregoscope)
P9	Objective 40x / 0.65	1	objective 40x / 0.65 (160/0.17, RMS thread), ⌀ 20 mm, 38 mm long (35 mm below the inner tube)	purchased
L1	Microscope slide	1	76 × 26 × 1 mm (microscopy standard)	purchased
D1	White 5 mm LED (diffused)	1	⌀ 5.0 mm; flange ⌀ 5.8 mm; height ≈ 8.6 mm; lead pitch 2.54 mm	purchased
R1	Resistor R1 (sets the current)	1	¼ W axial: body ≈ 6.3 × 2.3 mm; lead ⌀ 0.5 mm; pitch 10 mm	purchased
C1	Ceramic capacitor 100 nF	1	ceramic 100 nF (104) 50 V; pitch 5 mm	purchased
SW1	Rocker switch KCD1-101	1	panel cut-out 21.2 × 15.0 mm; 2 terminals	purchased
BT1	9 V battery (6F22) with snap clip	1	6F22: 48.5 × 26.5 × 17.5 mm	purchased
U1	LM317T adjustable regulator	1	TO-220: 10.0 × 4.5 × 15.0 mm; pins ADJ-OUT-IN, pitch 2.54 mm	purchased
PCB1	Driver board	1	1.6 mm thick; M3 holes 3.5 mm from the edges	purchased
H1	M3 × 10 self-tapping screw	4	M3 × 10; pan head ⌀ ≈ 5.6 mm	purchased
H2	M3 × 16 self-tapping screw	2	M3 × 16; pan head ⌀ ≈ 5.6 mm	purchased
H3	M3 × 8 self-tapping screw	4	M3 × 8; pan head ⌀ ≈ 5.6 mm	purchased

ID	Name	Qty	Standard size	Type
W1	Flexible wire 24 AWG	1	24 AWG: conductor ⌀ 0.51 mm; insulation ⌀ ≈ 1.4 mm	purchased
W2	Heat-shrink tube ⌀2-3 mm	1	⌀ 2 to 3 mm before shrinking; 10 mm pieces	purchased
W3	Hot glue and double-sided tape	1	glue stick ⌀ 7 mm; 12 mm double-sided tape	purchased

The sizes are the catalogue ones; check the real parts with jigs G1 to G3 before printing in series. Parts P1 to P10 are the pregoscope itself and are supplied by the customer.

4. Schematic and wiring



Soldering order on the board

1. Solder U1 (label facing forward: ADJ, OUT, IN), R1 (68 Ω) and C1 (100 nF) to the pads of the board. The traces of the board make the connections between them.
2. LED D1: orange wire (long leg, anode) to pad D1 ANODO; black wire (short leg, cathode) to pad D1 CATODO.
3. Battery BT1: red wire of the clip to pad BT1 POS; black wire to pad BT1 NEG.
4. Switch SW1: terminal 1 (red wire) to pad SW1 A; terminal 2 (orange wire) to pad SW1 B.
5. Cover the LED joints with heat-shrink and tie down loose wires with a dab of hot glue.

Circuit connections (reference for perforated board)

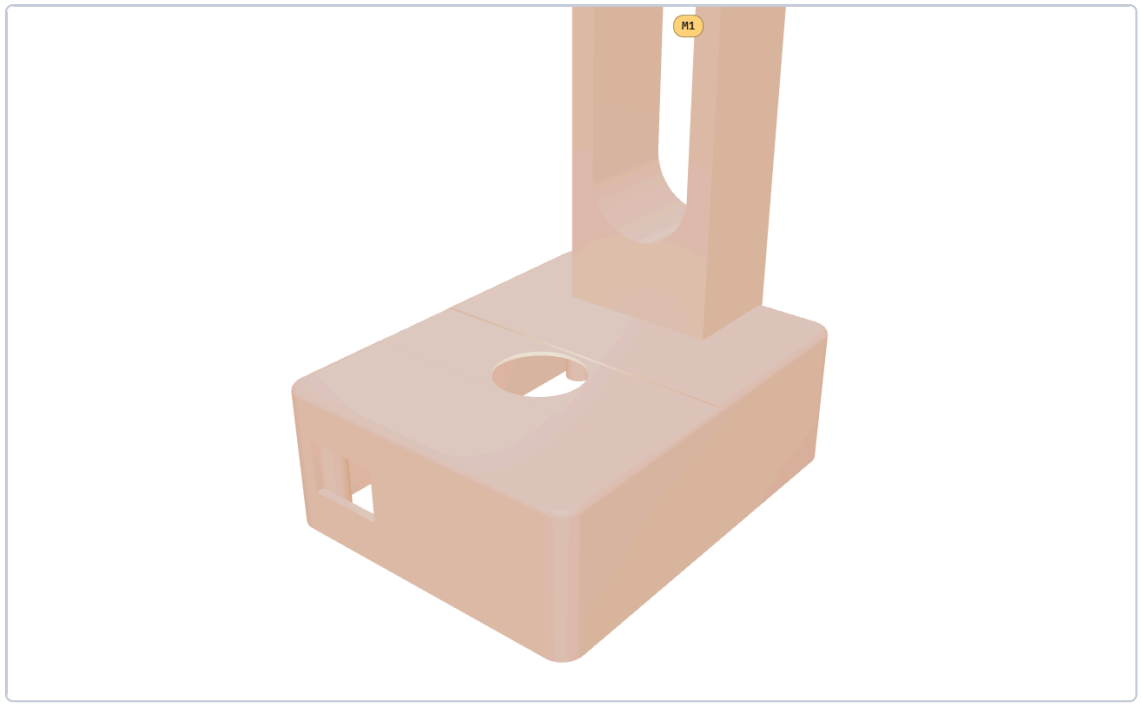
1. Red wire of clip BT1 to terminal 1 of switch SW1.
2. Terminal 2 of SW1 to pin IN of U1 (LM317T seen from the front, letters toward you: ADJ, OUT, IN from left to right).
3. C1 (100 nF) between pin IN of U1 and GND, soldered close to U1.
4. Pin OUT of U1 to one end of R1 (68 Ω).
5. The other end of R1 to pin ADJ of U1 and to the anode (long leg) of D1: all three at the same point.
6. Cathode (short leg) of D1 to the black wire of clip BT1 (GND). The other terminal of C1 also goes to GND.

Red wire = positive, black = GND. The LM317 holds 1.25 V across R1, so the LED current stays fixed.

1

Prepare the box (M1)

20 min



Remove the print supports and clean every hole with a 3 mm drill bit. Sand the edges of the slide channel lightly and make sure no debris is left inside the box.

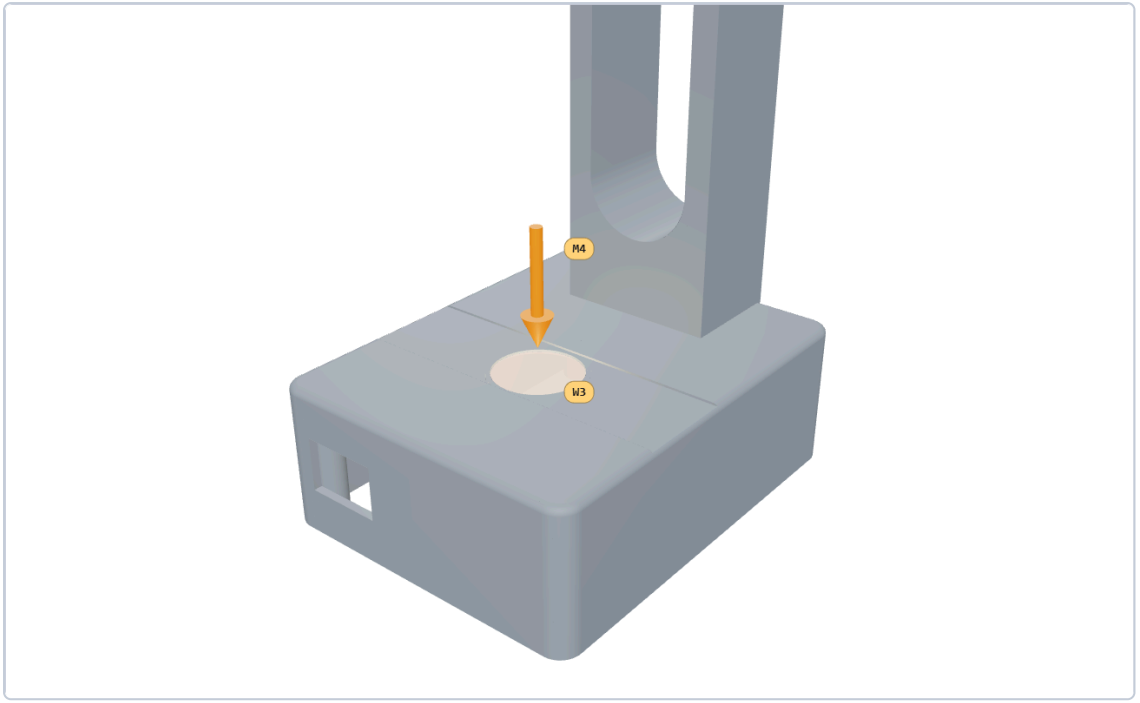
✓
The slide channel must be smooth and clean.

ID	Part in this step	Qty
M1	Support: light box, column and arm	1

2

Glue the diffuser (M4)

10 min



Place the disc M4 (⌀ 26 × 0.8 mm) in the ceiling seat from inside the box. Fix it with a thin bead of hot glue (W3) around the edge, without covering the ⌀ 24 mm window.

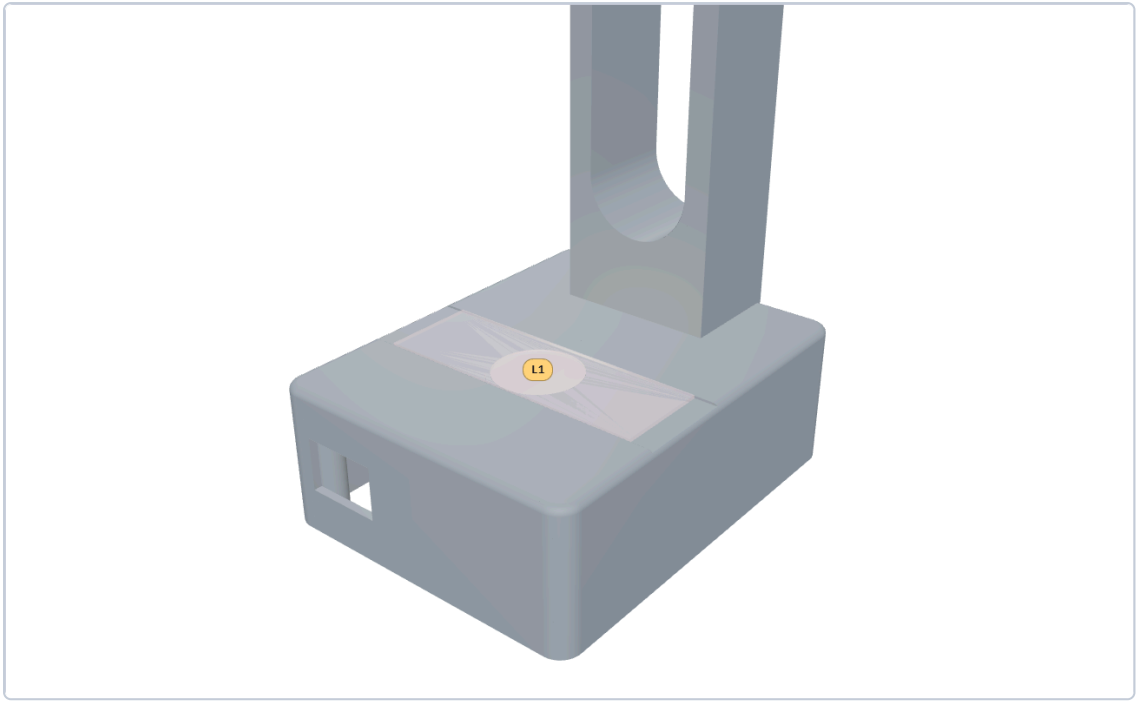
⚠
Hot glue burns: use the tip of the gun and let it cool for 1 minute.

ID	Part in this step	Qty
M4	Diffuser	1
W3	Hot glue and double-sided tape	1

3

Test-fit the slide (L1)

5 min



Slide the slide L1 (76 × 26 × 1 mm) through the channel. It must go in and out without effort and without visible side play. If it sticks, use jig G2 to pick the right size before printing again.

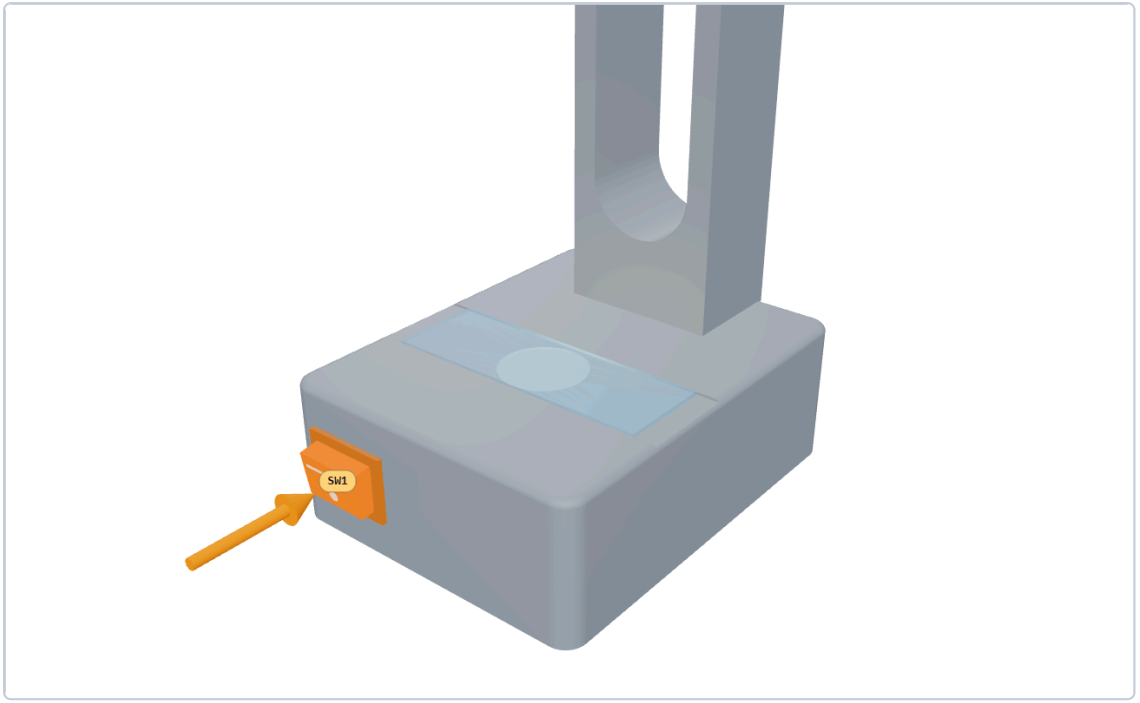
✓
The slide enters from the side and sits 40.4 mm above the floor.

ID	Part in this step	Qty
L1	Microscope slide	1

4

Install the switch (SW1)

5 min



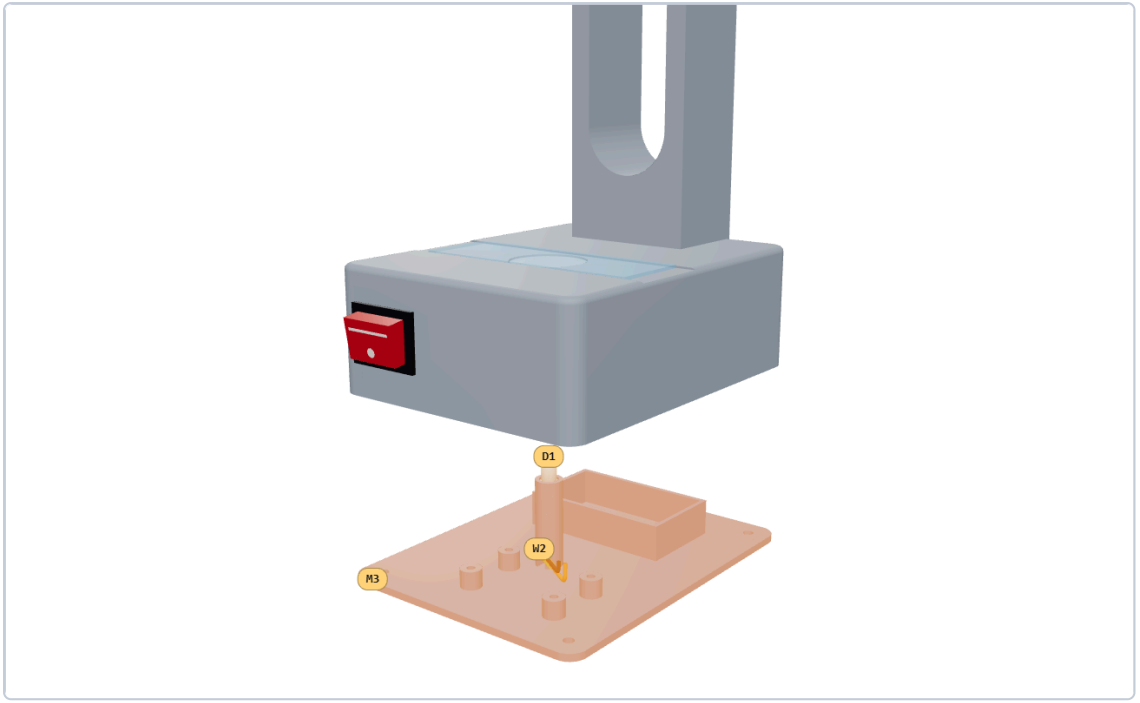
Push the KCD1-101 switch from outside into the 21.2 × 15.0 mm cut-out of the front wall until the tabs click. The two terminals end up inside the box.

ID	Part in this step	Qty
SW1	Rocker switch KCD1-101	1

5

Lid: mount the LED (M3, D1)

20 min



Solder two wires (W1) to the LED legs: orange to the long leg (anode), black to the short one (cathode). Insulate each joint with heat-shrink (W2). Seat the LED in the LED post of lid M3. The wires go down the hole of the post and leave through the side slot, which faces the board.

⚠
Solder each leg for less than 3 seconds so the LED does not burn out. Soldering iron at 300 °C: rest it in its stand and never touch the tip. Solder in a ventilated place.

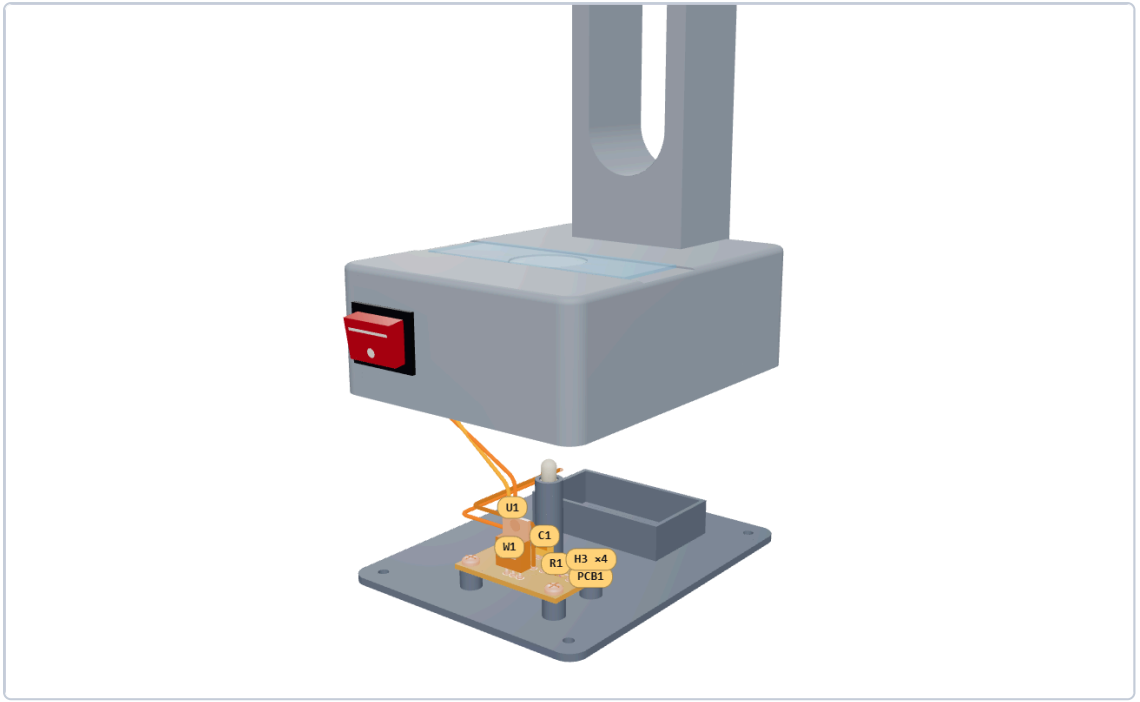
✓
Long leg = anode (+). The lid has no hole through to the outside.

ID	Part in this step	Qty
M3	Bottom lid	1
D1	White 5 mm LED (diffused)	1
W1	Flexible wire 24 AWG	1
W2	Heat-shrink tube ⌀2-3 mm	1

6

Lid: driver board (PCB1, U1, R1, C1)

40 min



Solder the regulator U1, the resistor R1 (68 Ω) and the capacitor C1 (100 nF) to PCB1 (40 × 25 mm), on the pads of the Gerber files. U1 faces forward with its label: ADJ, OUT, IN from left to right. Screw the board to the 4 standoffs of the lid with M3 × 8 (H3). Solder the wires of the LED and of the other parts to the marked pads, in the order of the wiring section.

⚠
Soldering iron at 300 °C: rest it in its stand and never touch the tip. Solder in a ventilated place.

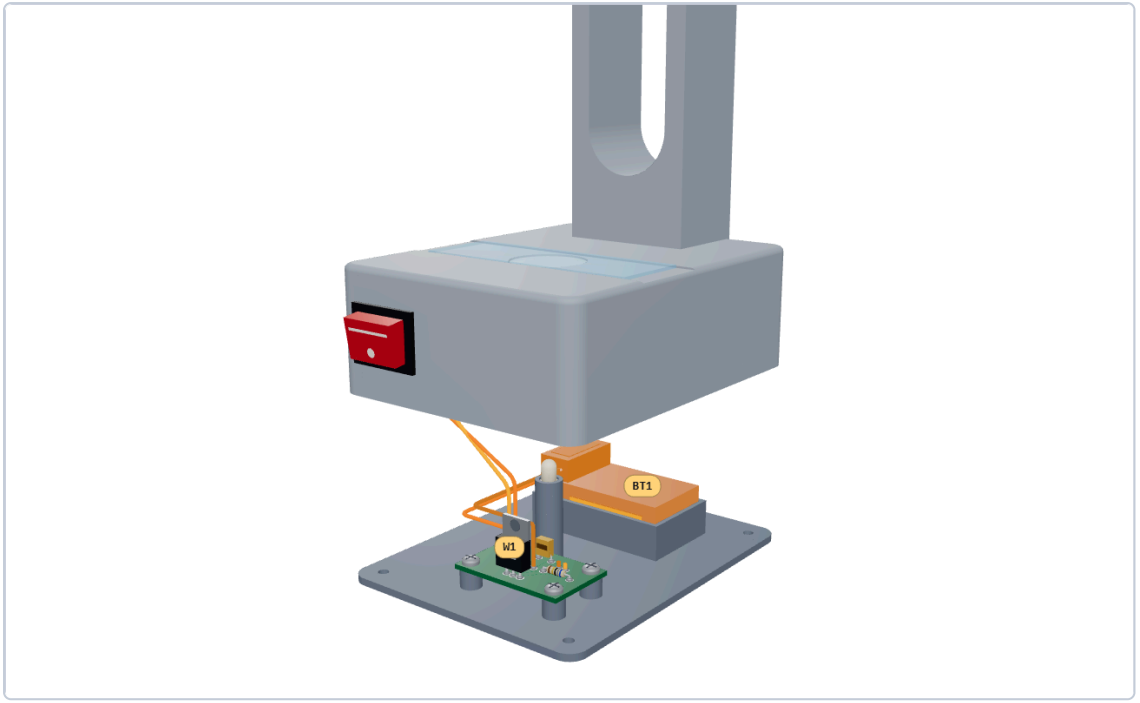
✓
Tab of U1 facing back; no solder joint touches another.

ID	Part in this step	Qty
PCB1	Driver board	1
U1	LM317T adjustable regulator	1
R1	Resistor R1 (sets the current)	1
C1	Ceramic capacitor 100 nF	1
H3	M3 × 8 self-tapping screw	4

7

Lid: 9 V battery (BT1)

10 min



Seat the 9 V battery in the cradle of the lid. Solder the red wire of the clip to the BT1 POS pad and the black one to the BT1 NEG pad, and only then snap the clip onto the battery. The wires of the switch SW1 go to the pads SW1 A and B.

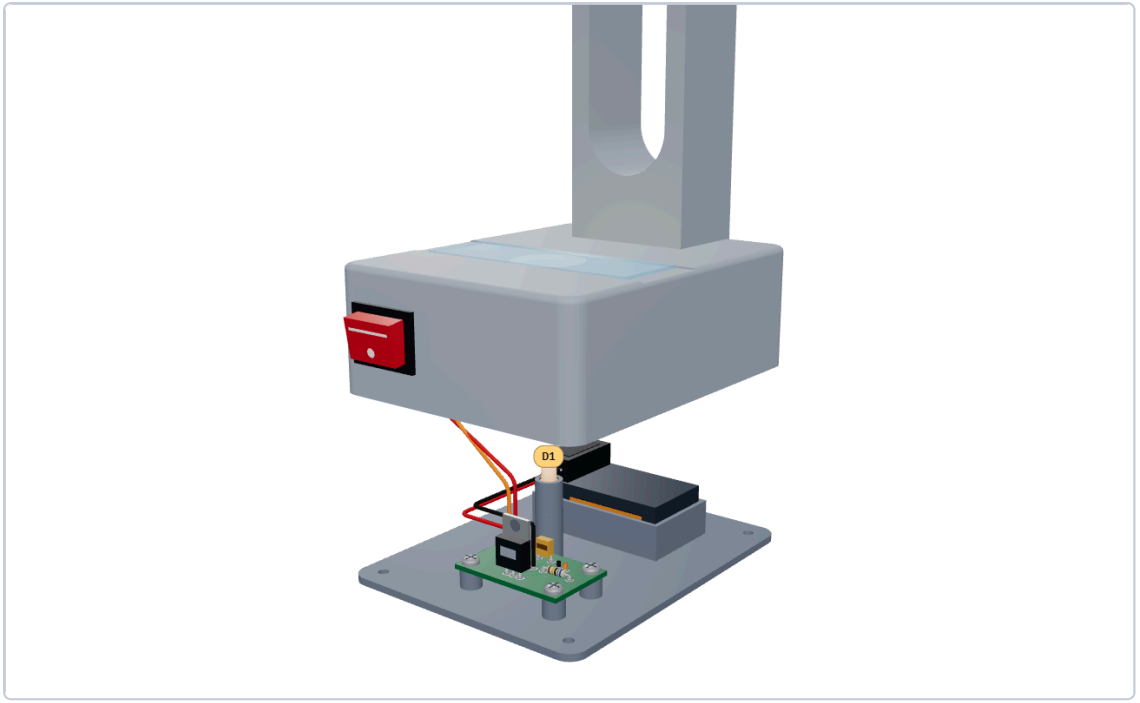
⚠
Check the polarity before powering up: LED (long leg +), diode (band = cathode), plug with positive centre.

ID	Part in this step	Qty
BT1	9 V battery (6F22) with snap clip	1

8

Electrical test with the multimeter

15 min



Before closing, switch the power on and follow the test plan of the electrical test section point by point. Every reading must be inside its range and the LED must light. If a reading is out of range, stop and fix it before going on.

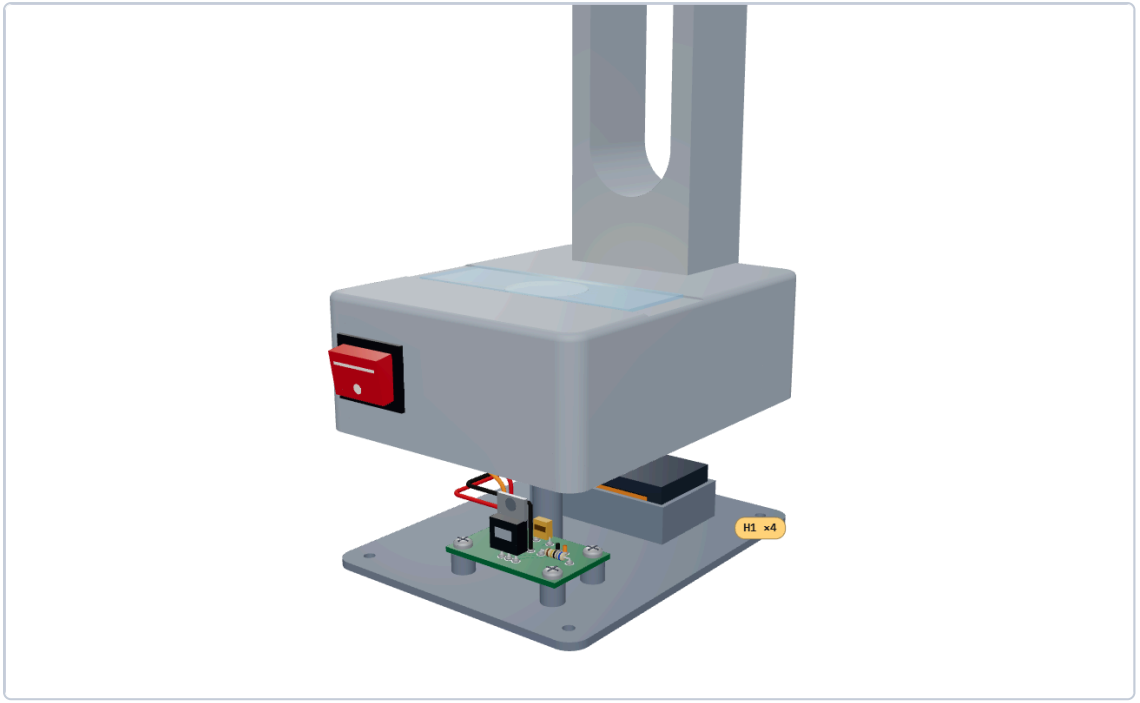
⚠
Check the polarity before powering up: LED (long leg +), diode (band = cathode), plug with positive centre.

✓
18.4 mA in the LED

9

Close the lid (H1)

10 min



Raise the lid to the box without trapping any wire and tighten the 4 screws M3 × 10 (H1) at the corners. Tighten a little at a time, in a cross pattern, without forcing the plastic.

⚠
Do not overtighten: PLA cracks. Stop when the screw seats.

ID	Part in this step	Qty
H1	M3 × 10 self-tapping screw	4

10

Fit the pregoscope (M2, H2, P1 to P10)

15 min



Take off the removable half of the clamp M2, lean the outer tube P2 into the column ring, put the half back and tighten the 2 screws M3 × 16 (H2) without forcing. The head P1 rests on top of the ring and the objective P9 sits over the window.

⚠
Do not overtighten: PLA cracks. Stop when the screw seats.

ID	Part in this step	Qty
M2	Tube clamp	1
H2	M3 × 16 self-tapping screw	2
P1	Head: plate and threaded rod	1
P2	Outer tube	1
P3	Inner tube (telescopic)	1
P4	Left jaw	1
P5	Right jaw	1

ID	Part in this step	Qty
P6	Right knob	1
P10	Left knob (mirror of P6)	1
P9	Objective 40x / 0.65	1

11

Focus and use

10 min



Switch SW1 on. Loosen the clamp and slide the outer tube, or pull the inner tube P3, until the slide is sharp. Nominal focus: the lens 0.8 mm from the slide, with the inner tube pulled out 19 mm (read the scale printed on the tube, at the edge of the outer tube). The tip of the objective is spring-loaded, but do not push it against the glass.

6. Electrical test

Supply: a new 9 V battery, white LED of 3.1 V. Measure in this order.

Measuring point	Expected	Range	If out of range	Measured
Battery BT1: positive to negative	9.00 V	8.4 to 9.7 V	Weak battery, clip not seated or connection reversed.	
Input of U1 (after SW1) to GND	9.00 V	8.3 to 9.7 V	Switch off, cold joint at SW1 or a loose wire.	
Output of U1 (OUT) to GND	4.35 V	4.1 to 4.6 V	R1 open or LED reversed: the output rises close to the input.	
Anode of D1 (pin ADJ) to GND	3.10 V	2.8 to 3.4 V	LED reversed (reads 0 V) or burnt out (reads above 3.4 V).	
Across R1 (between OUT and ADJ)	1.25 V	1.2 to 1.3 V	Outside 1.25 V the LM317 is fake or out of regulation.	
LED current, multimeter in series	18.4 mA	17.4 to 19.4 mA	Wrong R1 (check the 68 Ω bands: blue-grey-black) or LM317 not regulating.	

If a reading is out of range, stop and fix it before closing the box.

7. Calibration: jigs G1 to G3

Print the jigs before the final parts, with the same settings, lying flat and without supports. Try the real part in each hole, from the smallest to the largest, and pick the smallest one where it goes in and slides without too much play.

G1 — Tube jig (calibracao-tubo.stl)

Series	Holes (mm)	Parameter to update
`P2` Outer tube	`30.2 · 33.0 · 33.4 · 33.8 · 34.2 · 34.6 · 36.4`	tubeDiameter = chosen value − 0.2

G2 — Parts jig (calibracao-pecas.stl)

Series	Holes (mm)	Parameter to update
`L1` Slide (1.2 mm recess)	`26.2 · 26.4 · 26.6 · 26.8 · 27.0`	slideWidth = chosen value − 0.6
`SW1` Switch KCD1-101	`20.8×14,6 · 21.0×14,8 · 21.2×15,0 · 21.4×15,2 · 21.6×15,4`	switchCutout (width and height)
`D1` LED flange	`5.8 · 6.0 · 6.2 · 6.4 · 6.6`	ledFlange = chosen value − 0.4
`H1` M3 pilot hole	`2.4 · 2.6 · 2.8`	pilotHole (tight fit that still threads)
`J1` Jack DC-022B	`7.6 · 7.8 · 8.0 · 8.2`	jackDiameter
`RV1` Rheostat M7 thread	`7.0 · 7.3 · 7.5 · 7.8`	knobDiameter

G3 — Battery jig (calibracao-bateria.stl)		
Series	**Holes (mm)**	**Parameter to update**
BT1 9 V battery	48.6×26,6 · 49.1×27,1 · 49.6×27,6 · 50.1×28,1	batteryFit = chosen value − battery measurement

After choosing the values, update the parameters in holder.ts, regenerate the files and issue revision B.

8. Final inspection

#	Check	OK	Not OK
1	The slide L1 moves in the channel without effort and without visible side play.	☐	☐
2	The pregoscope tube fits the ring and is held by M2 without cracking the PLA.	☐	☐
3	With the inner tube pulled out 19 mm on its scale the lens is about 0.8 mm from the slide and the image focuses.	☐	☐
4	The LED lights and the light is even over the window, with no hot spot.	☐	☐
5	LED current measured inside the range of the electrical test.	☐	☐
6	Lid closed with 4 screws, no trapped wire and no gap larger than 0.5 mm.	☐	☐
7	The switch snaps in; on v2 the knob turns smoothly and the plug seats.	☐	☐
8	Nothing loose when shaken gently; no burrs on the printed parts.	☐	☐

Assembler		**Date**		**Inspector**	